

OSNOVE STROJNOG UČENJA
Laboratorijske vježbe
Završni zadatak

POPUNJAVA STUDENT:

Datum:

Ime i prezime:

LV grupa:

Napomena: Rješenja se predaju u PDF formatu. Ostali formati NEĆE biti ocijenjeni.

Sve priložene slike moraju biti generirane na temelju pripadajućeg programskog koda.

1. Zadatak

a. –

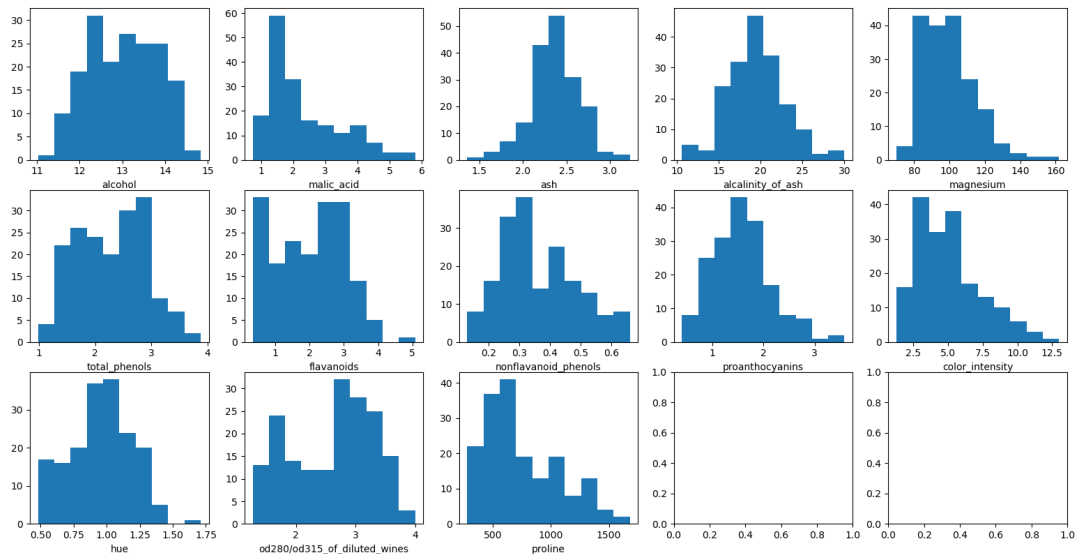
b. Zalijepite sliku iz terminala koja prikazuje osnovnu statistiku podatkovnog skupa.

```
Broj uzoraka: 178
Broj značajki: 13
Značajke: Index(['alcohol', 'malic_acid', 'ash', 'alcalinity_of_ash', 'magnesium',
                'total_phenols', 'flavanoids', 'nonflavanoid_phenols',
                'proanthocyanins', 'color_intensity', 'hue',
                'od280/od315_of_diluted_wines', 'proline'],
              dtype=object)
Izlazna veličina: target
Moguće vrijednosti: [0 1 2]
count    alcohol  malic_acid    ash  alcalinity_of_ash  magnesium  ...  color_intensity    hue  od280/od315_of_diluted_wines    proline    target
mean    13.000618    2.336348    2.366517    19.494944    99.741573  ...    5.058090    0.957449    2.611685    746.893258    0.938202
std     0.811827    1.117146    0.274344    3.339564    14.282484  ...    2.318286    0.228572    0.709990    314.907474    0.775035
min     11.030000    0.740000    1.360000    10.600000    70.000000  ...    1.280000    0.480000    1.270000    278.000000    0.000000
25%     12.362500    1.602500    2.210000    17.200000    88.000000  ...    3.220000    0.782500    1.937500    500.500000    0.000000
50%     13.050000    1.865000    2.360000    19.500000    98.000000  ...    4.690000    0.965000    2.780000    673.500000    1.000000
75%     13.677500    3.082500    2.557500    21.500000    107.000000  ...    6.200000    1.120000    3.170000    985.000000    2.000000
max     14.830000    5.800000    3.230000    30.000000    162.000000  ...    13.000000    1.710000    4.000000    1680.000000    2.000000

[8 rows x 14 columns]
```

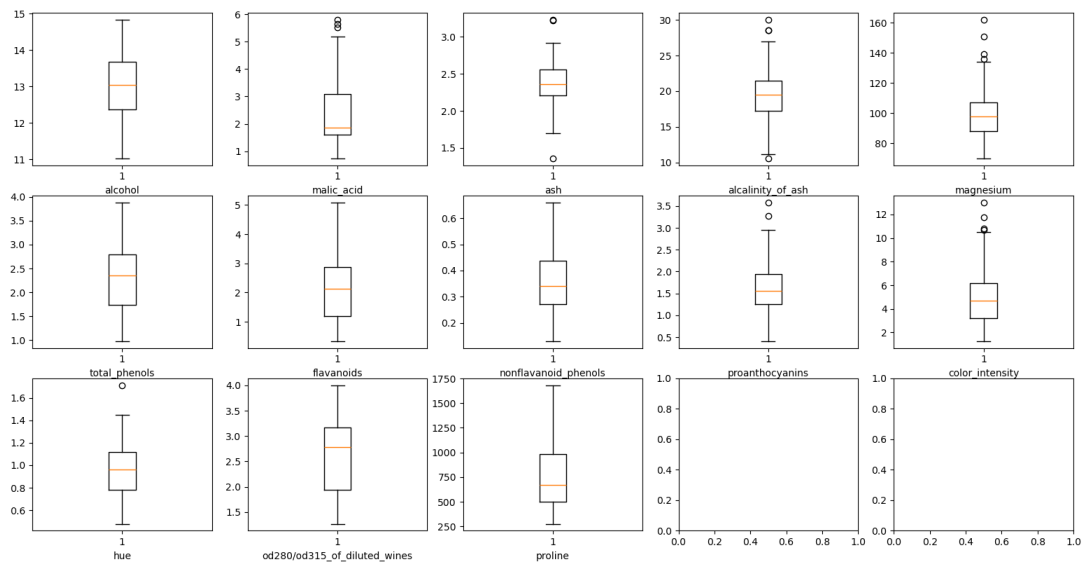
c. Zalijepite sliku histograma.

Histogram svih značajki



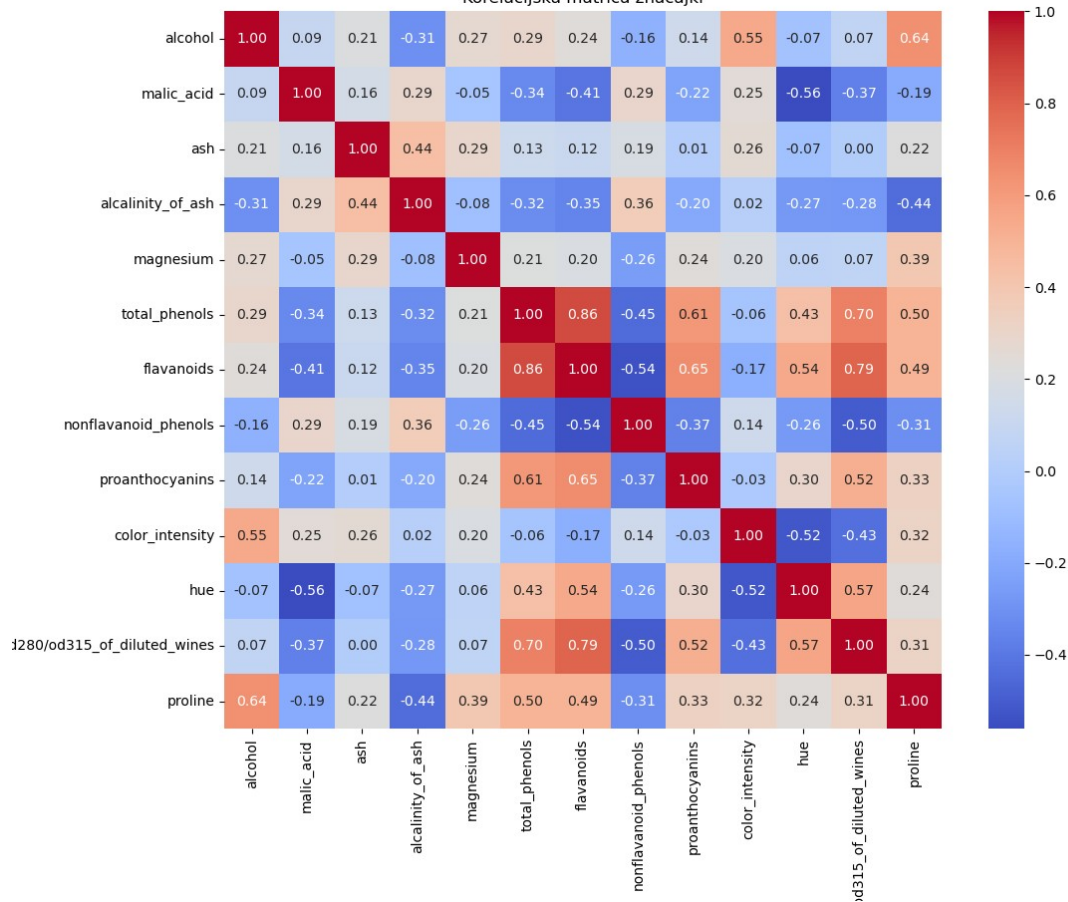
d. Zalijepite sliku kutijastih dijagrama.

Kutijasti dijagram svih značajki



e. Zalijepite sliku korelacijske matrice (*heat mapa*).

Korelacijska matrica značajki



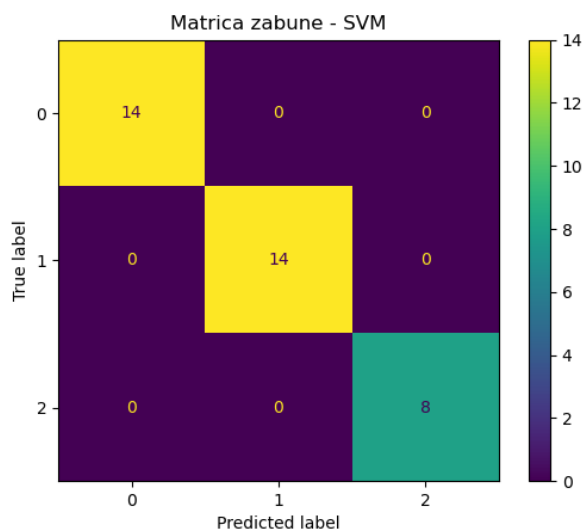
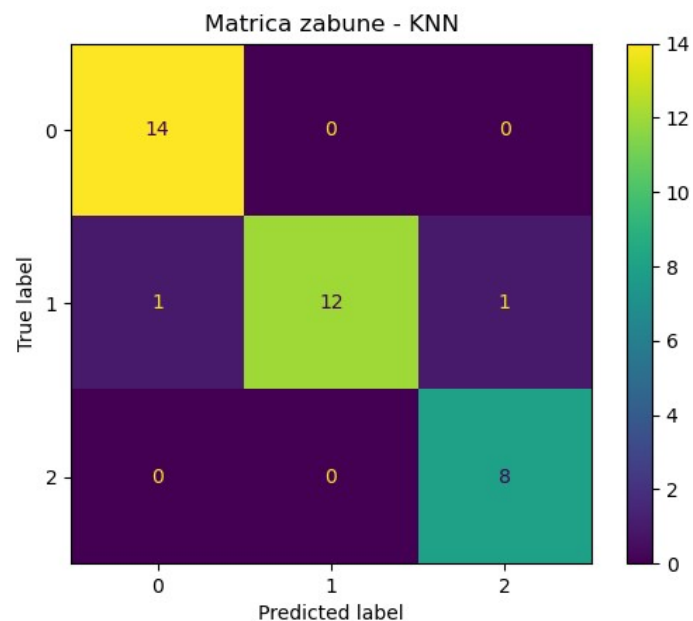
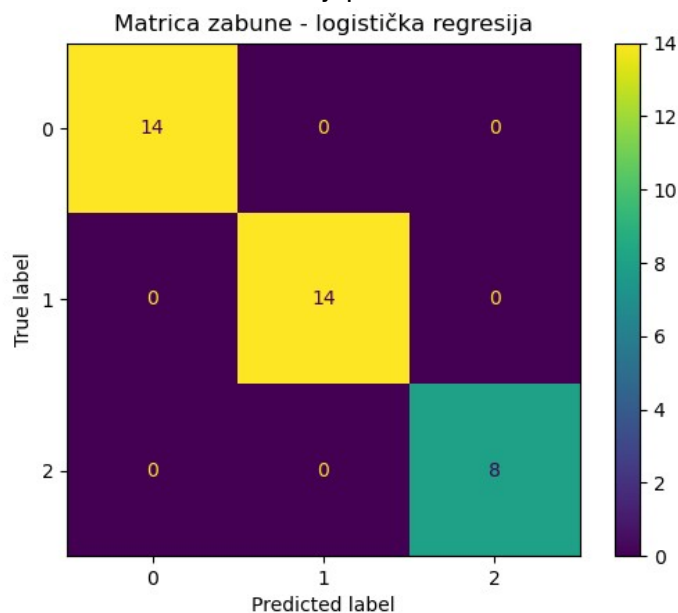
2. Zadatak

a. —

b. —

c. —

d. Zalijepite slike matrica zabune na testnom skupu za sve modele (ukupno 4 slike):



e. —

f. —

g. Zalijepite sliku grafa metode lakta:



3. Zadatak

a) Zalijepite *screenshot* s informacijama o mreži iz terminala:

Model: "sequential"

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 16)	224
dense_1 (Dense)	(None, 8)	136
dense_2 (Dense)	(None, 3)	27

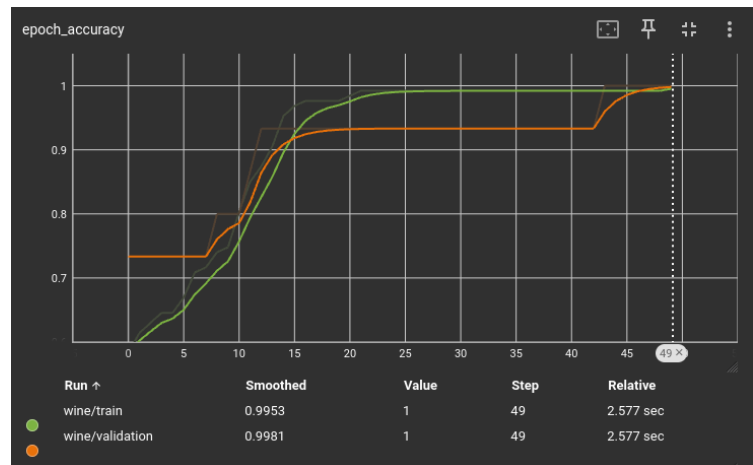
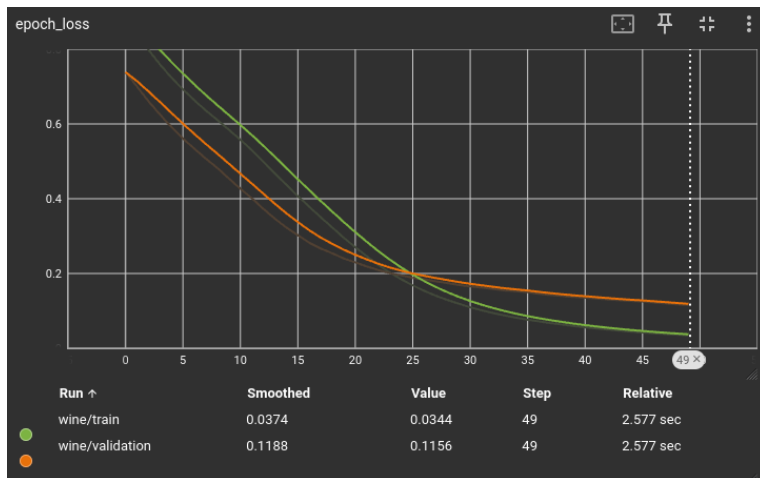
Total params: 387 (1.51 KB)
Trainable params: 387 (1.51 KB)
Non-trainable params: 0 (0.00 B)

b) –

c) –

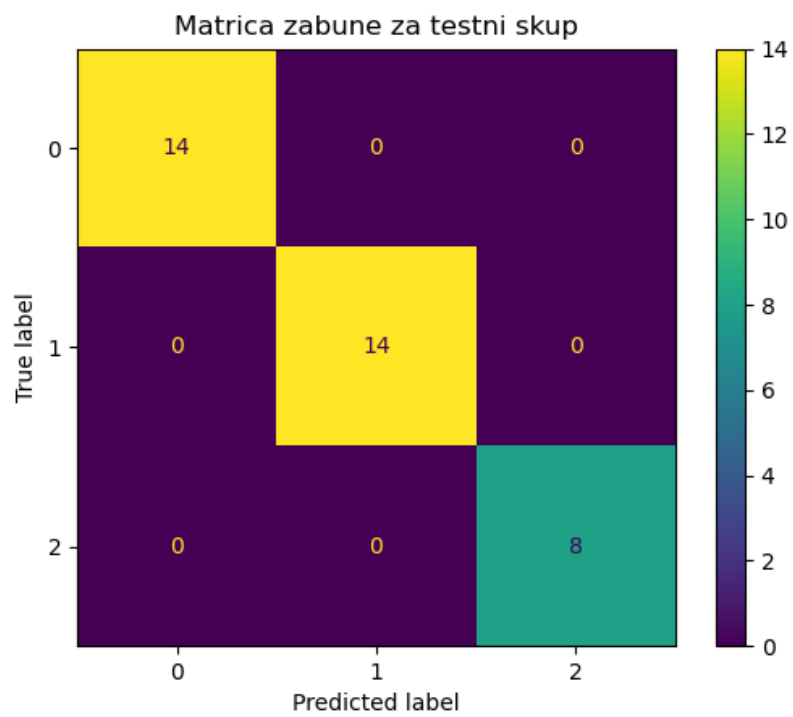
d) –

e) Zalijepite slike iz *Tensorboarda* na kojima se vidi napredak učenja kroz epohe po parametrima *loss* i *accuracy*:



f) –

g) Zalijepite sliku matrice zabune:



h)